msvtx_el2hcal

MC vs data · overview · rank 45 / 98

rank 45 / 98 by endcap D_equal

used by: NN1 barrel, NN1 endcap

rank-identical columns merged into this one:
    summedCaloClusters/el2hcal
    MS1Vtx_l2hcal_raw
    MS1Vtx_barrel_l2hcal
    MS1Vtx_endcap_l2hcal

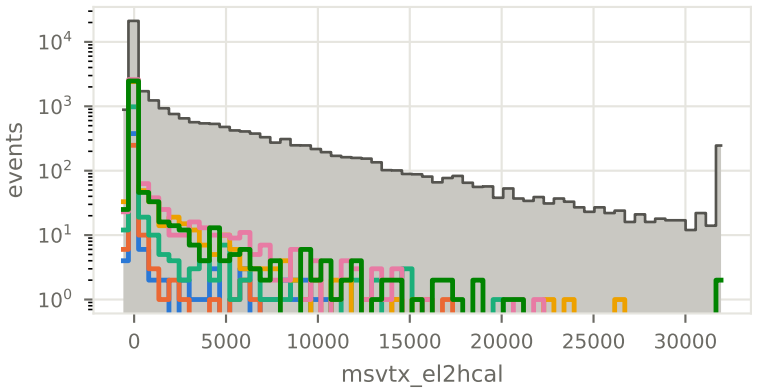
region	D_equal	noise	D_pooled	125/15	125/55	200/50	400/100	600/150	1000/275	n_bkg	rank
endcap	0.311	0.022	0.309	0.323	0.310	0.316	0.311	0.299	0.312	34,970	45/98
barrel	0.259	0.051	0.269	0.293	0.239	0.246	0.283	0.277	0.267	1,089	51/98
all	0.296	0.018	0.297	0.316	0.274	0.295	0.303	0.292	0.297	36,059	49/98

msvtx_el2hcal

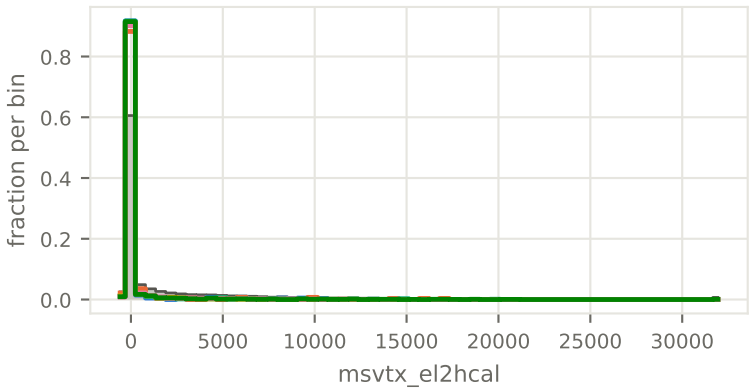
region: ENDCAP · rank 45 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.311 (noise 0.022) D_pooled 0.309 min 0.299 (mH600_mS150) max 0.323 (mH125_mS15) bkg n = 34,970 sig n = 9,996

1a · each mass point vs background · raw

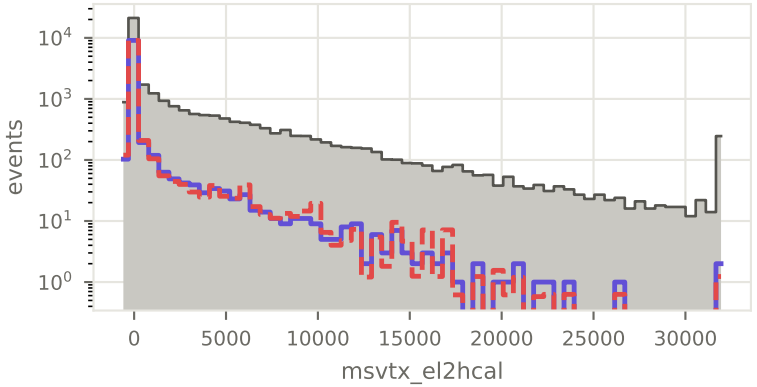


1b · each mass point vs background · normalized

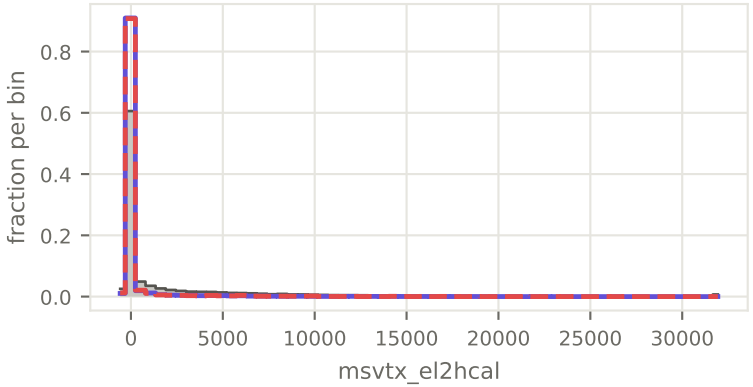


1 · each mass point
background n 34,970
mH125_mS15 D 0.323 n 411
mH125_mS55 D 0.310 n 281
mH200_mS50 D 0.316 n 1,071
mH400_mS100 D 0.311 n 2,648
mH600_mS150 D 0.299 n 2,882
mH1000_mS275 D 0.312 n 2,703

2a · combined signal vs background · raw

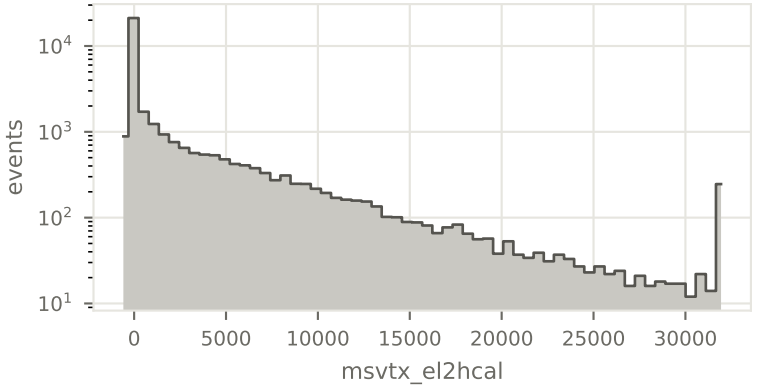


2b · combined signal vs background · normalized

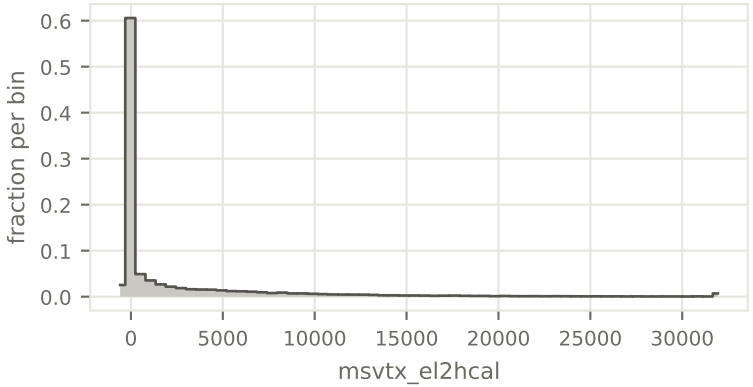


2 · combined signal
background n 34,970
combined, pooled D 0.309 n 9,996
equal weight per mass D 0.311

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 34,970

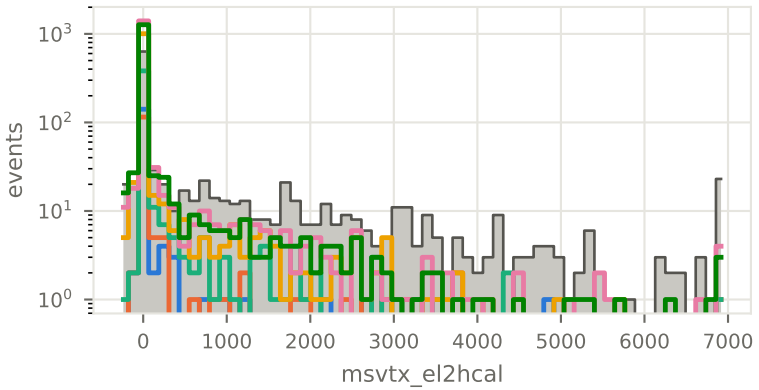
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el2hcal

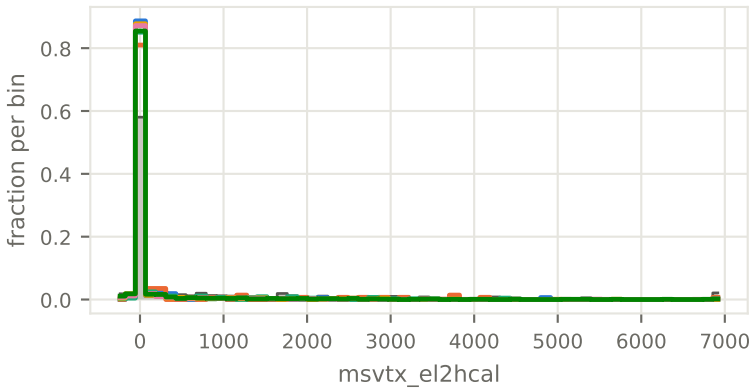
region: BARREL · rank 45 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.259 (noise 0.051) D_pooled 0.269 min 0.239 (mH125_mS55) max 0.293 (mH125_mS15) bkg n = 1,089 sig n = 4,987

1a · each mass point vs background · raw

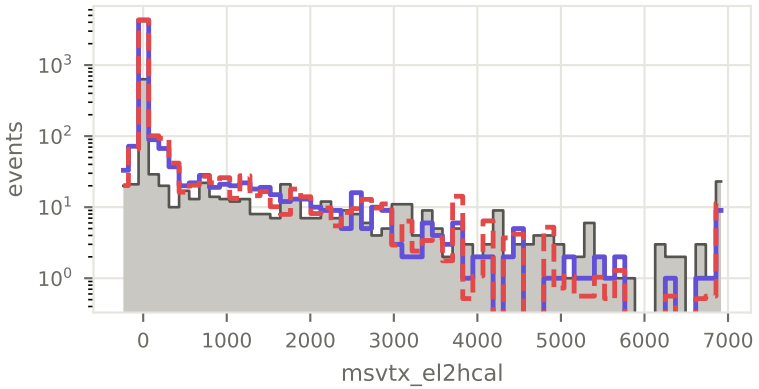


1b · each mass point vs background · normalized

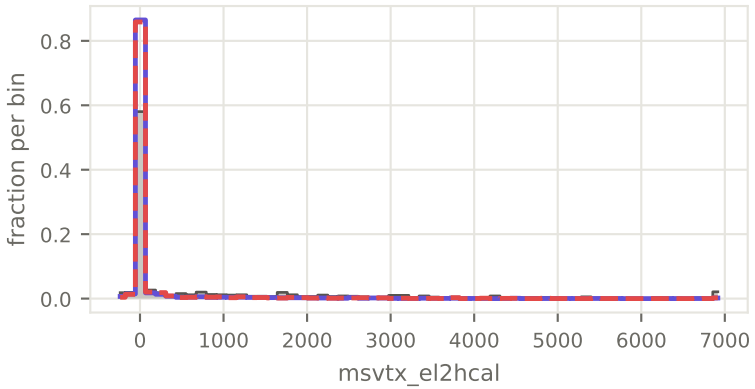


1 · each mass point
background n 1,089
mH125_mS15 D 0.293 n 159
mH125_mS55 D 0.239 n 142
mH200_mS50 D 0.246 n 449
mH400_mS100 D 0.283 n 1,144
mH600_mS150 D 0.277 n 1,608
mH1000_mS275 D 0.267 n 1,485

2a · combined signal vs background · raw

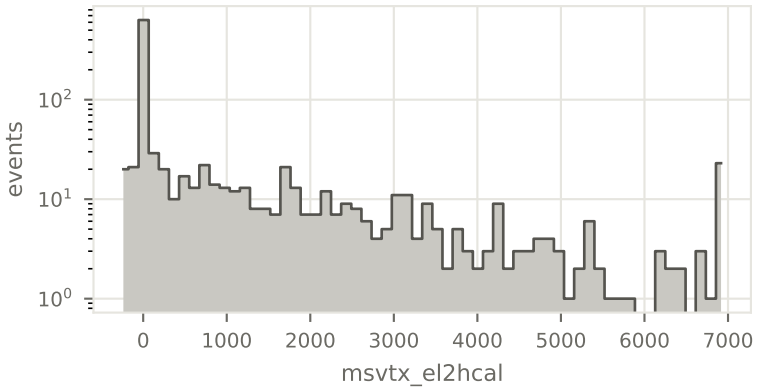


2b · combined signal vs background · normalized

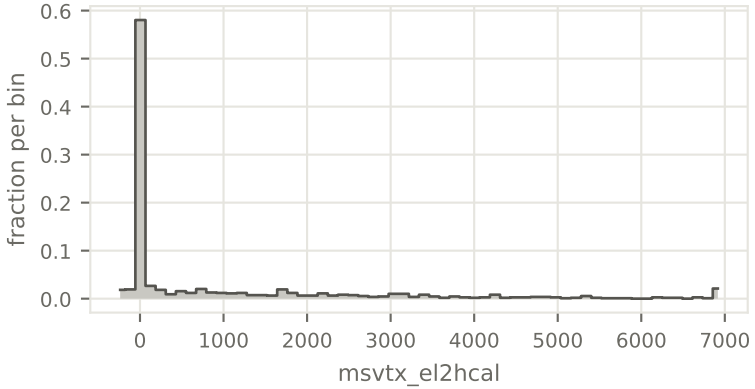


2 · combined signal
background n 1,089
combined, pooled D 0.269 n 4,987
equal weight per mass D 0.259

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 1,089

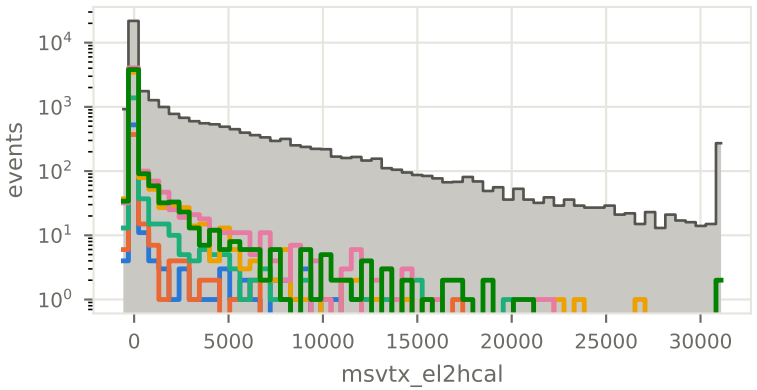
a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed

msvtx_el2hcal

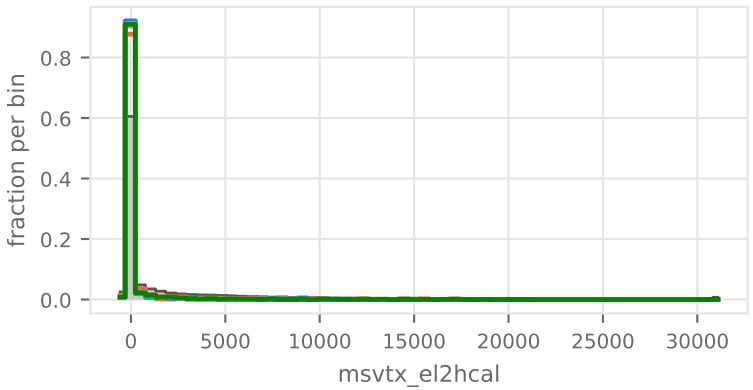
region: ALL · rank 45 / 98 · in: NN1 barrel, NN1 endcap

D_equal 0.296 (noise 0.018) D_pooled 0.297 min 0.274 (mH125_mS55) max 0.316 (mH125_mS15) bkg n = 36,059 sig n = 14,983

1a · each mass point vs background · raw

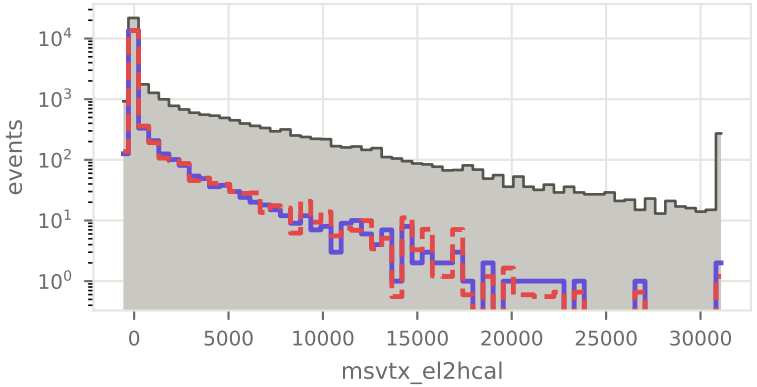


1b · each mass point vs background · normalized

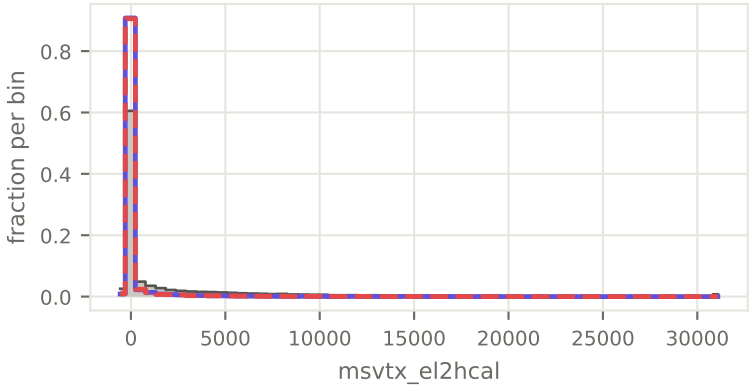


1 · each mass point
background n 36,059
mH125_mS15 D 0.316 n 570
mH125_mS55 D 0.274 n 423
mH200_mS50 D 0.295 n 1,520
mH400_mS100 D 0.303 n 3,792
mH600_mS150 D 0.292 n 4,490
mH1000_mS275 D 0.297 n 4,188

2a · combined signal vs background · raw

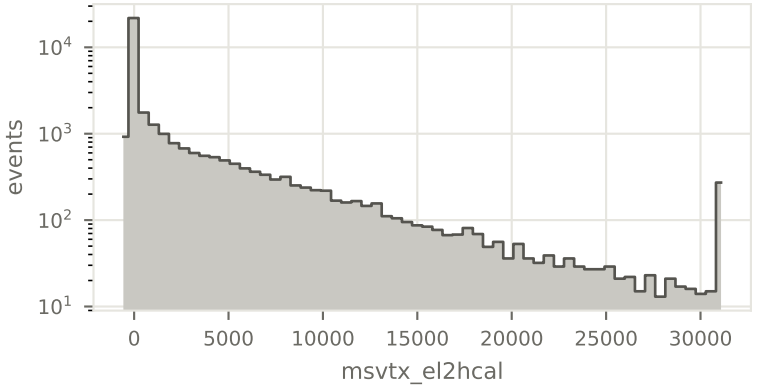


2b · combined signal vs background · normalized

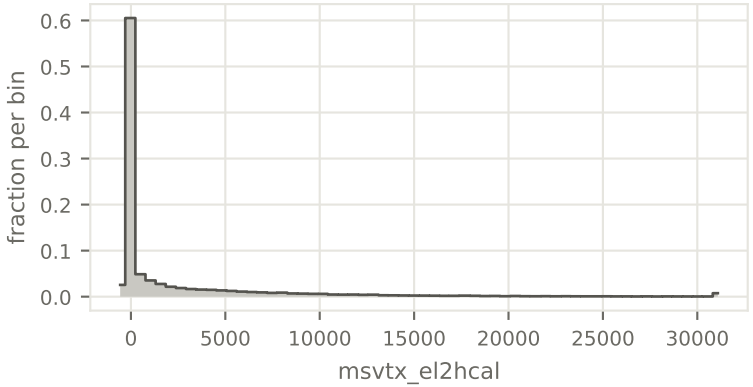


2 · combined signal
background n 36,059
combined, pooled D 0.297 n 14,983
equal weight per mass D 0.296

3a · background alone · raw



3b · background alone · normalized



3 · background alone
NaN 0.0% −999 0.0%
background n 36,059

a = raw (log y) b = normalized · edge bins include under/overflow · NaN and -999 removed